Calibration of Probabilities

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Abstract

This project studies the calibration of probabilistic classifiers. We evaluate Logistic Regression and Random Forest, apply Platt scaling and Isotonic regression to improve probability estimates. The models are tested on two real-world datasets: Adult and Breast Cancer.

1 Introduction

In supervised learning, many probabilistic classifier models output probabilities rather than hard decisions, giving us a value between 0 and 1 representing the estimated likelihood of belonging to a certain class.

However, since these probabilities are not always reliable, we introduce calibration that refers to the ability of a model to produce reliable probability estimates. A model may achieve high classification accuracy while still providing poorly calibrated probability estimates. It measures how much these probabilities correspond to real statistical frequency of the events in the real world. High accuracy does not necessarily imply good calibration, since calibration concerns the quality of predicted probabilities rather than classification correctness [1].

In this project we study the calibration of probabilistic classifiers, considering two models, Logistic Regression and Random Forest, trained using scikit-learn. We then apply post-processing calibration methods, Platt scaling and Isotonic regression, both implemented from scratch.

The performance will be evaluated using Accuracy, Log-loss, Brier score and with representation of Reliability diagrams.

2 Datasets

We used two real-world binary classification datasets, one obtained from OpenML and the other from the `sklearn.datasets` library. The two datasets were intentionally selected with different characteristics in order to analyze how calibration methods behave under different data conditions.

The first dataset is the Adult dataset, based on the American census of 1994, containing a mix of numerical and categorical data, with the objective of predicting if a person earns more than 50k dollars per year. This dataset contains a relatively large number of samples and is more balanced compared to the second dataset.

The second dataset used in this project is the Breast Cancer Wisconsin dataset. It is a binary classification dataset containing numerical features computed from digitized images of breast mass cell nuclei. The objective is to classify tumors as malignant or benign. Compared to the Adult dataset, this dataset is significantly smaller, containing 569 samples. This characteristic makes it particularly useful for analyzing the behavior of calibration methods in settings with limited calibration data.

3 Methodology

3.1 Data preprocessing

Both datasets were preprocessed before training. Numerical features were standardized using a standard scaler, which transforms the data to have zero mean and unit variance. Categorical features were converted into binary columns using one-hot encoding, configured to ignore unknown categories during inference.

To ensure a robust evaluation and prevent data leakage, each dataset was split into training (60%), validation (20%), and test (20%) sets using stratified sampling to maintain the original class distribution. This specific three-way split is crucial for calibration: the training set was exclusively used to fit the base models (Logistic Regression and Random Forest), while the validation set was strictly reserved for fitting the post-processing calibration methods (Platt scaling and Isotonic regression). Finally, the test set was used solely for evaluating the true generalization and calibration performance of the models. A model is perfectly calibrated if:

$$P(Y = 1 \mid \hat{P} = p) = p$$

Table 1: Summary and structural properties of the benchmark datasets.

Dataset Property / Metric	**Adult (Census Income)**	**Breast Cancer Wisconsin**
Total Samples (N)	$48,842$	569
Original Features	14 (6 Num, 8 Cat)	30 (30 Num, 0 Cat)
Processed Features (Post-OHE)	100+	30
Missing Values (NaNs)	$6,465$	0
Target Mapping (y)	$>$ 50K $\rightarrow$ 1, $\leq$ 50K $\rightarrow$ 0	Benign $\rightarrow$ 1, Malignant $\rightarrow$ 0
Positive Class Ratio ($y = 1$)	23.93%	62.74%
Split Ratios (Train / Val / Test)	60%/20%/20%	60%/20%/20%
Sample Sizes (Train / Val / Test)	$29,305\,/\,9,768\,/\,9,769$	$341\,/\,114\,/\,114$

3.2 Models

We implemented two classification models:

- **Logistic Regression**: a linear model that outputs probabilities through a sigmoid function.

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

  A key theoretical property of Logistic Regression is that its cost function directly optimizes the log-likelihood of the data, which is mathematically equivalent to minimizing the Log Loss (cross-entropy). Because Log Loss is a strictly proper scoring rule, the model is inherently driven to produce well-calibrated probability estimates by design.

- **Random Forest**: an ensemble method composed of multiple decision trees trained on bootstrap samples of the data. Each tree is built using a random subset of features, and predictions are obtained by majority voting. Probabilities are estimated as the average of predictions across trees.

3.3 Calibration methods

To improve the quality of probability estimates, we applied two post-processing calibration techniques:

- **Platt scaling**: a method that fits a logistic regression model on the predicted probabilities of the base model, transforming them using a sigmoid function.

$$P(y = 1 \mid x) = \frac{1}{1 + \exp(Af(x) + B)}$$

- **Isotonic regression**: a non-parametric method that fits a monotonic function to map predicted probabilities to calibrated values. In this project, isotonic regression was implemented using the Pool Adjacent Violators (PAV) algorithm. It starts with one block for each validation example and repeatedly merges adjacent blocks whenever their average labels violate the monotonicity constraint.

Both calibration models were trained using the validation set.

3.4 Evaluation metrics

We evaluated model performance using the following metrics:

- **Accuracy**: measures the proportion of correctly classified samples.
- **Log Loss**: evaluates the quality of predicted probabilities, penalizing confident but incorrect predictions.

$$\text{LogLoss} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right]$$

- **Brier Score**: bounded in [0, 1], measures the mean squared error between predicted probabilities and true labels.

$$\text{Brier} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2$$

In addition, we used **reliability diagrams** to visually assess calibration. These plots compare predicted probabilities with the observed frequency of the positive class. A perfectly calibrated model corresponds to a diagonal line in the reliability diagram.

4 Results

4.1 Adult Dataset

Model	Accuracy	Log Loss	Brier
LR	0.8556	0.3152	0.1007
LR + Platt	0.8555	0.3311	0.1031
LR + Iso	0.8528	0.3285	0.1015
RF	0.8622	0.3161	0.0991
RF + Platt	0.8619	0.3134	0.0980
RF + Iso	0.8620	0.3129	0.0967

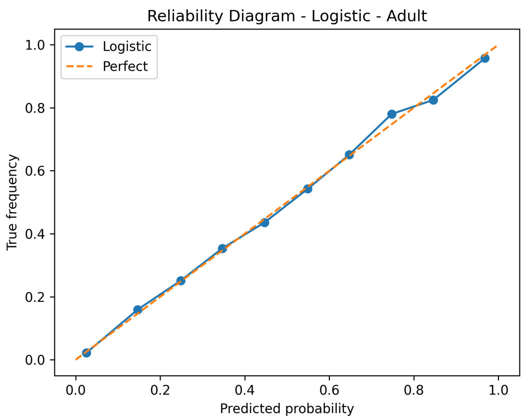
Table 2: Results on the Adult dataset.

4.2 Breast Cancer Dataset

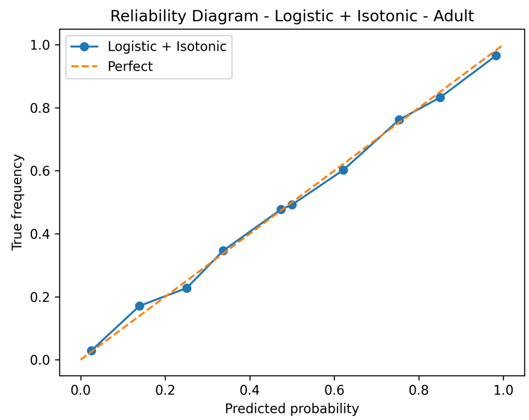
Model	Accuracy	Log Loss	Brier
LR	0.9912	0.0719	0.0190
LR + Platt	0.9912	0.1397	0.0254
LR + Iso	0.9561	1.5149	0.0439
RF	0.9386	0.1361	0.0434
RF + Platt	0.9474	0.2016	0.0489
RF + Iso	0.9298	0.1381	0.0483

Table 3: Results on the Breast Cancer dataset.

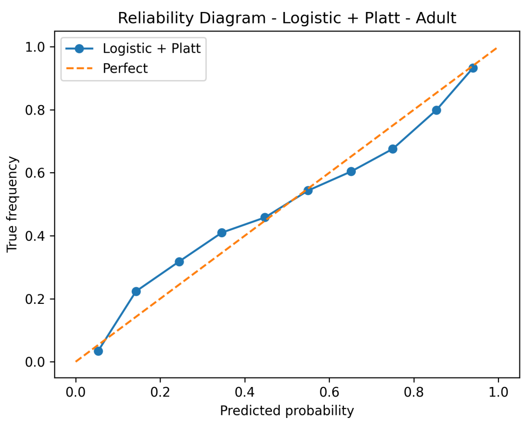
4.3 Reliability Diagrams



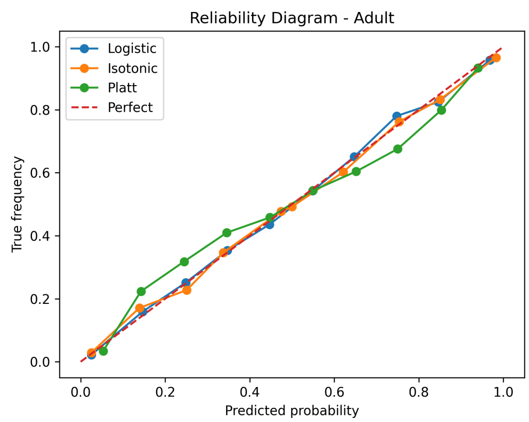
(a) Logistic Regression



(b) Logistic + Isotonic

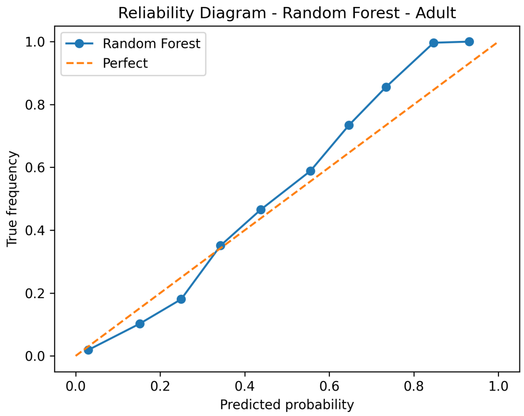


(c) Logistic + Platt

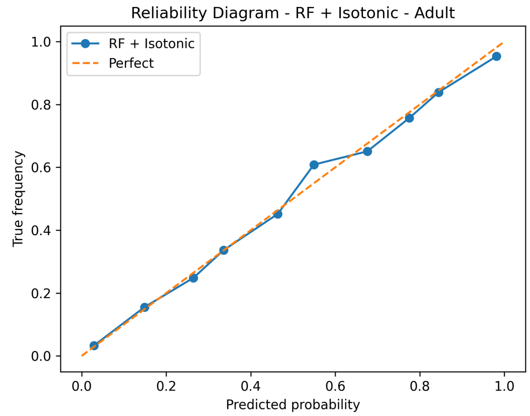


(d) Comparison

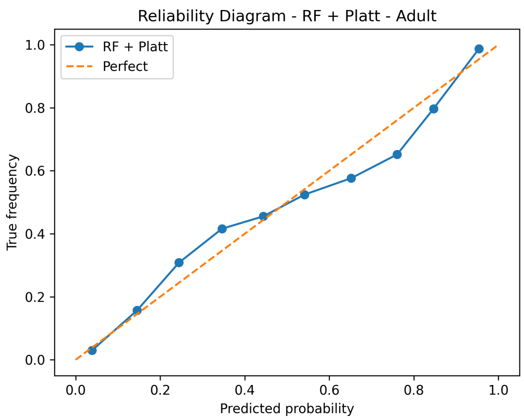
Figure 1: Calibration curves for Logistic Regression on the Adult dataset.



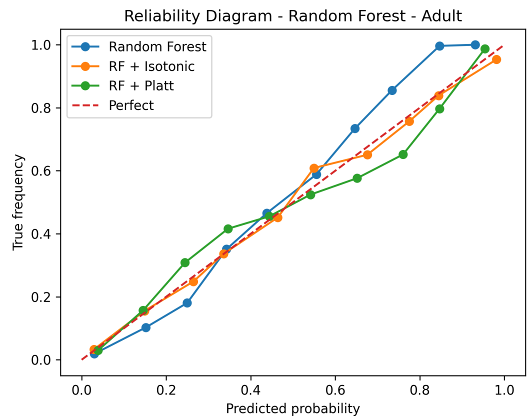
(a) Random Forest



(b) RF + Isotonic

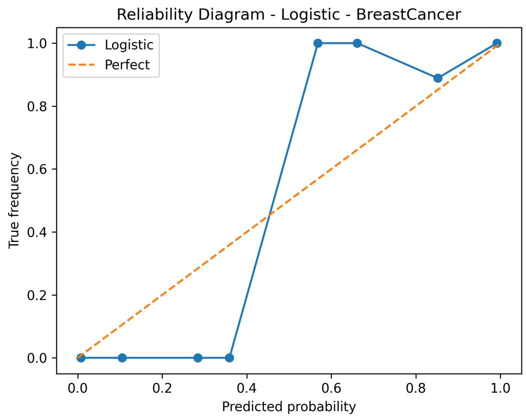


(c) RF + Platt

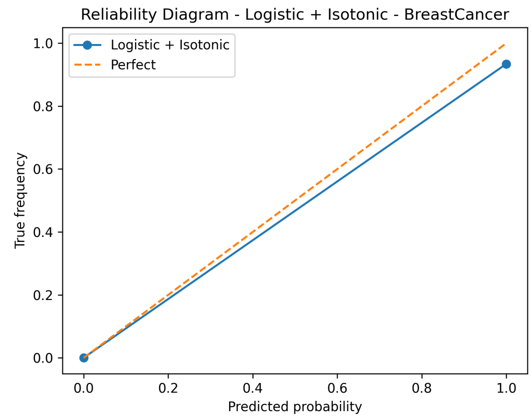


(d) Comparison

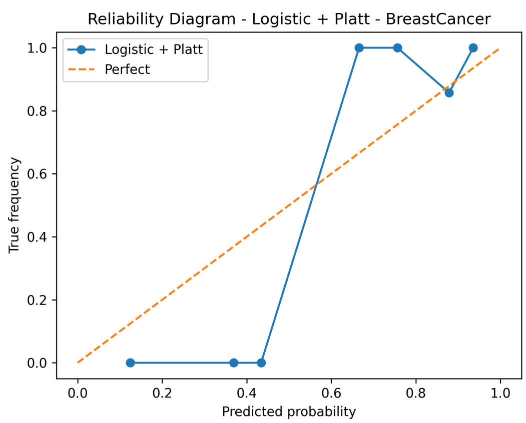
Figure 2: Calibration curves for Random Forest on the Adult dataset.



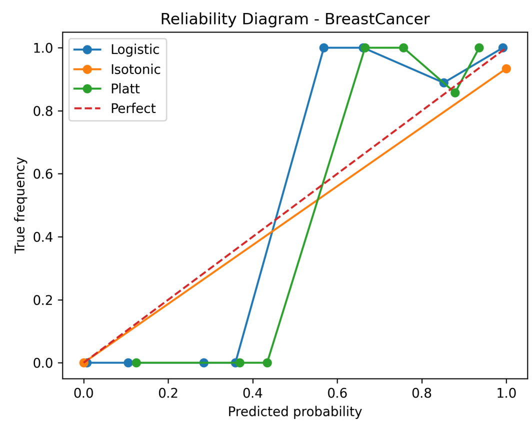
(a) Logistic Regression



(b) Logistic + Isotonic

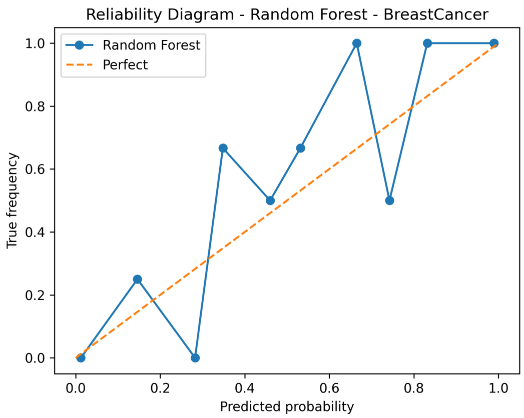


(c) Logistic + Platt

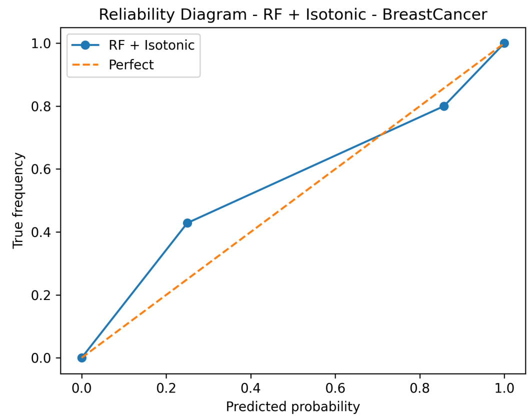


(d) Comparison

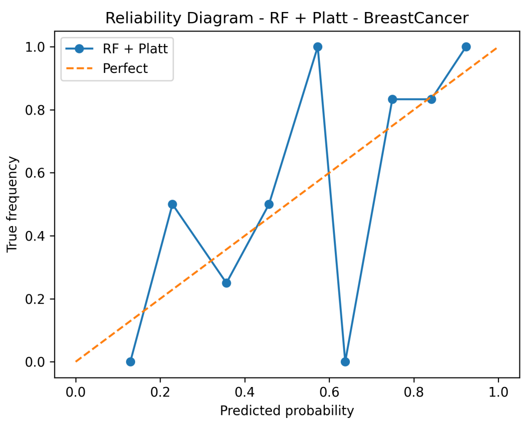
Figure 3: Calibration curves for Logistic Regression on the Breast Cancer dataset.



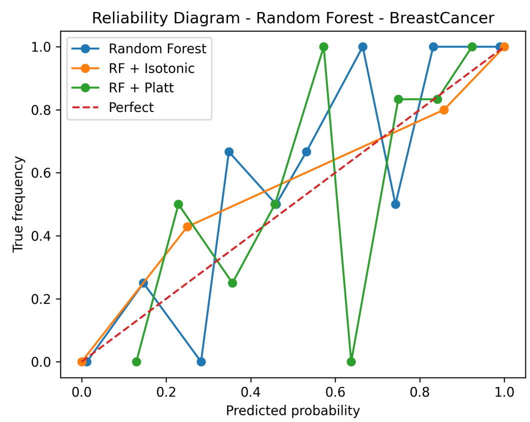
(a) Random Forest



(b) RF + Isotonic



(c) RF + Platt



(d) Comparison

Figure 4: Calibration curves for Random Forest on the Breast Cancer dataset.

5 Analysis

The empirical evaluation highlights significant interactions between the base classifier architecture, post-hoc calibration strategies, and calibration dataset capacity.

Logistic Regression Base Calibration: Across both datasets, raw Logistic Regression exhibits naturally well-calibrated probabilities (Log Loss of 0.3152 on Adult and 0.0719 on Breast Cancer). This aligns with theoretical expectations: because Logistic Regression directly optimizes binary Cross-Entropy (Log Loss), which is a strictly proper scoring rule, its uncalibrated probability predictions already match empirical likelihoods closely [1]. Consequently, applying Platt scaling or Isotonic regression to Logistic Regression yields minimal gains or slightly inflates Log Loss (e.g., rising to 0.3311 with Platt on Adult), as post-processing introduces minor estimation noise onto an already optimal parametric curve [2].

Random Forest Dynamics & Platt Scaling Success: For Random Forest, probability estimates are obtained by averaging the leaf-node class frequencies across all trees in the ensemble, rather than through hard majority voting. The number of trees (numbers of estimators) directly controls the stability of this average: with few trees, individual probability estimates carry high variance, and by chance some predictions land near 0 or 1 even for genuinely ambiguous cases. Because Log Loss penalizes confident-but-wrong predictions very heavily, this variance-induced overconfidence — not a bias toward intermediate values — is what produced the inflated Log Loss (0.7134) observed on the small Breast Cancer test set with a low tree count. Increasing number of estimators stabilizes the per-tree average and substantially reduces this effect: in a controlled comparison, raising the ensemble from 10 to 200 trees left test accuracy unchanged (0.9386) while Log Loss dropped from 0.4330 to 0.1361, confirming that the improvement operates purely at the level of probability quality rather than classification correctness. Applying Platt scaling to the 10-tree Random Forest on Breast Cancer provides a similar correction by re-scaling the ensemble's raw probabilities through a fitted sigmoid, but this benefit diminishes once the ensemble itself is stabilized with more trees, with number estimators = 200, the uncalibrated Random Forest already outperforms both Platt- and Isotonic-calibrated versions on this dataset, indicating that post-hoc calibration and ensemble size address the same underlying issue and are partially redundant.

Isotonic Regression and Dataset Scale Sensitivity: The experimental findings clearly highlight the sensitivity of non-parametric calibration to dataset size:

- On the large Adult dataset ($Validation\, N = 9,768$), Isotonic regression performs effectively. Combined with Random Forest, Isotonic calibration yields the top overall performance in Accuracy (0.8620) and Brier Score (0.0967).
- Conversely, on the small Breast Cancer dataset ($Validation\, N = 114$), Isotonic regression suffers from severe overfitting. Specifically, for Logistic Regression + Isotonic, Log Loss surges to 1.5149. Because Isotonic regression uses step-wise isotonic fitting (PAV algorithm), limited calibration samples lead to coarse step boundaries that heavily penalize test instances near probability extremes [1, 2].

Sampling Variance in Reliability Diagrams: A further consideration concerns the visual interpretation of the reliability diagrams themselves. On the Breast Cancer dataset, several diagrams (Figures 3 and 4) display pronounced zig-zag patterns even for models with high aggregate accuracy. This is a direct consequence of the test set size: with only 114 test samples distributed across 10 bins, individual bins contain as few as 5–15 points. This yields a 95% confidence interval as wide as $[0.42, 0.98]$. This confirms that binned reliability diagrams require substantially larger test sets than aggregate metrics such as accuracy or Log Loss to be visually reliable, and that the irregular shape of the Breast Cancer diagrams should not, by itself, be interpreted as evidence of poor calibration.

In summary, parametric methods like Platt scaling are highly robust on smaller datasets, whereas non-parametric methods like Isotonic regression require large validation sets to prevent overfitting and achieve optimal probability calibration. Similarly, visual calibration assessment via reliability diagrams requires sufficiently large test sets to be trustworthy, independent of the calibration method used.

6 Conclusion

In this project, we studied the calibration of probabilistic classifiers using Logistic Regression and Random Forest on two real-world datasets with different characteristics.

The results show that Logistic Regression generally provides reasonably calibrated probabilities by design, while Random Forest produces competitive accuracy but benefits from calibration when tree ensembles compress predicted distributions.

Among the post-processing calibration methods, Platt scaling proved to be remarkably stable and effective on small datasets, drastically lowering Log Loss for Random Forest on Breast Cancer. Isotonic regression proved highly sensitive to sample sizes: while achieving state-of-the-art results on large data (Adult dataset), it suffered from severe overfitting on small validation samples (Breast Cancer dataset).

Overall, our findings confirm the importance of evaluating not only classification accuracy, but also the reliability of predicted probabilities, selecting post-hoc calibration techniques tailored to dataset size and model architecture.

7 Acknowledgements

The work was inspired by the paper "Predicting Good Probabilities with Supervised Learning" by Niculescu-Mizil and Caruana, as well as the blog post "Understanding Model Calibration - A Gentle Introduction", which provided useful insights into calibration techniques and their interpretation.

References

[1] M. Pavlovic, *Understanding Model Calibration: A Gentle Introduction and Visual Exploration of Calibration and the Expected Calibration Error (ECE)*, ICLR Blogposts, 2025.

[2] A. Niculescu-Mizil, R. Caruana, *Predicting Good Probabilities With Supervised Learning*, Proceedings of the 22nd International Conference on Machine Learning (ICML), 2005.